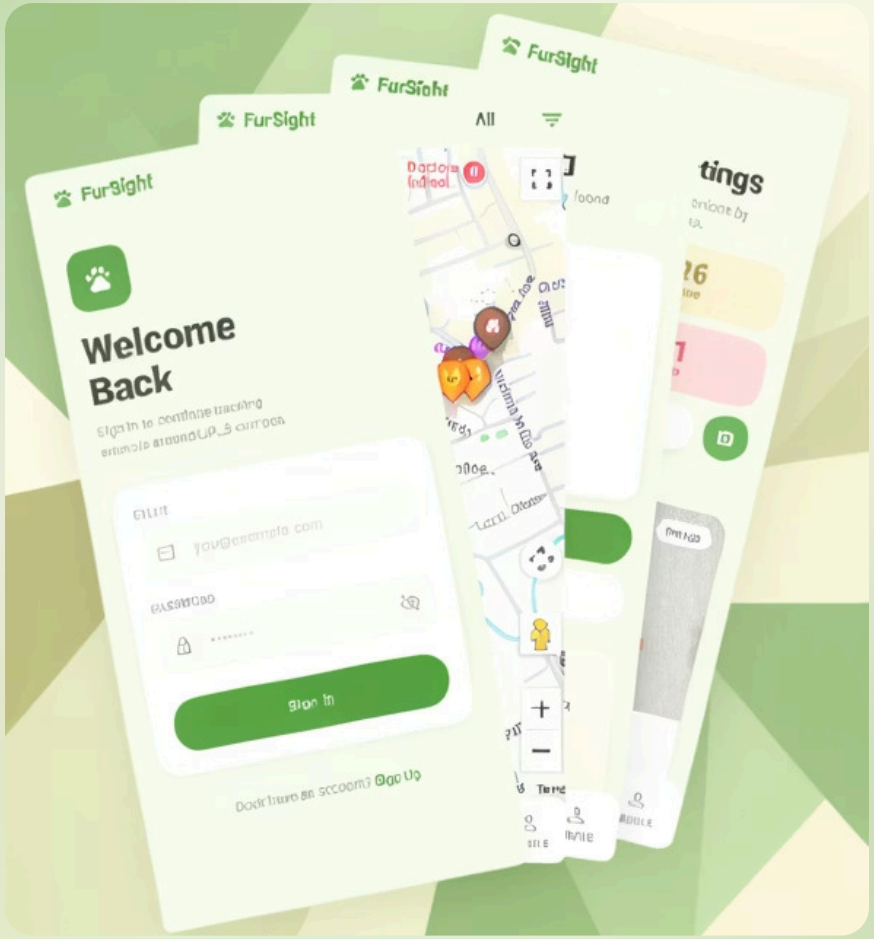

FurSight

Mobile app for geotagged sighting and AI-based identification of campus cats and dogs.

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Problem & Objectives

The Gap

UPLB has no centralized system to document or track its resident campus animals – sightings are anecdotal and undocumented.

Why it Matters

Without data, it's hard to monitor animal welfare, understand movement patterns, or even locate a missing campus animal.

Objectives

- Implement AI-based detection and individual re-identification from user-submitted photos
- Record geotagged sightings in a centralized Firebase database
- Visualize sighting data via a real-time heatmap

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Related Work & Model Choices

Community Tracking

Citizen science platforms (e.g., iNaturalist) show that geotagged community submissions can effectively map animal distributions at low cost.

Detection: YOLOv8n

YOLO architectures offer real-time object detection on mobile devices. YOLOv8n was selected for its speed-accuracy balance on our campus cat/dog dataset.

Re-identification

Deep metric learning with cosine similarity outperforms Siamese networks when per-individual training data is limited. Threshold calibration is critical.

Similarity Thresholds

Literature shows optimal thresholds balance precision and recall. Higher thresholds reduce false positives but risk missing valid matches. Our 0.55 threshold was calibrated empirically.

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Existing Apps & Gaps

App	What it does	What's missing
iNaturalist	Species ID, geotagged sightings	No individual re-ID, not campus-specific
PawBoost	Lost pet reporting	No AI, no heatmap, no re-ID
Finding Rover	Dog facial recognition	Dogs only, not open community-driven

Bottom line: No existing app combines community geotagged sighting + individual re-ID + campus-specific heatmap in one on-device pipeline.

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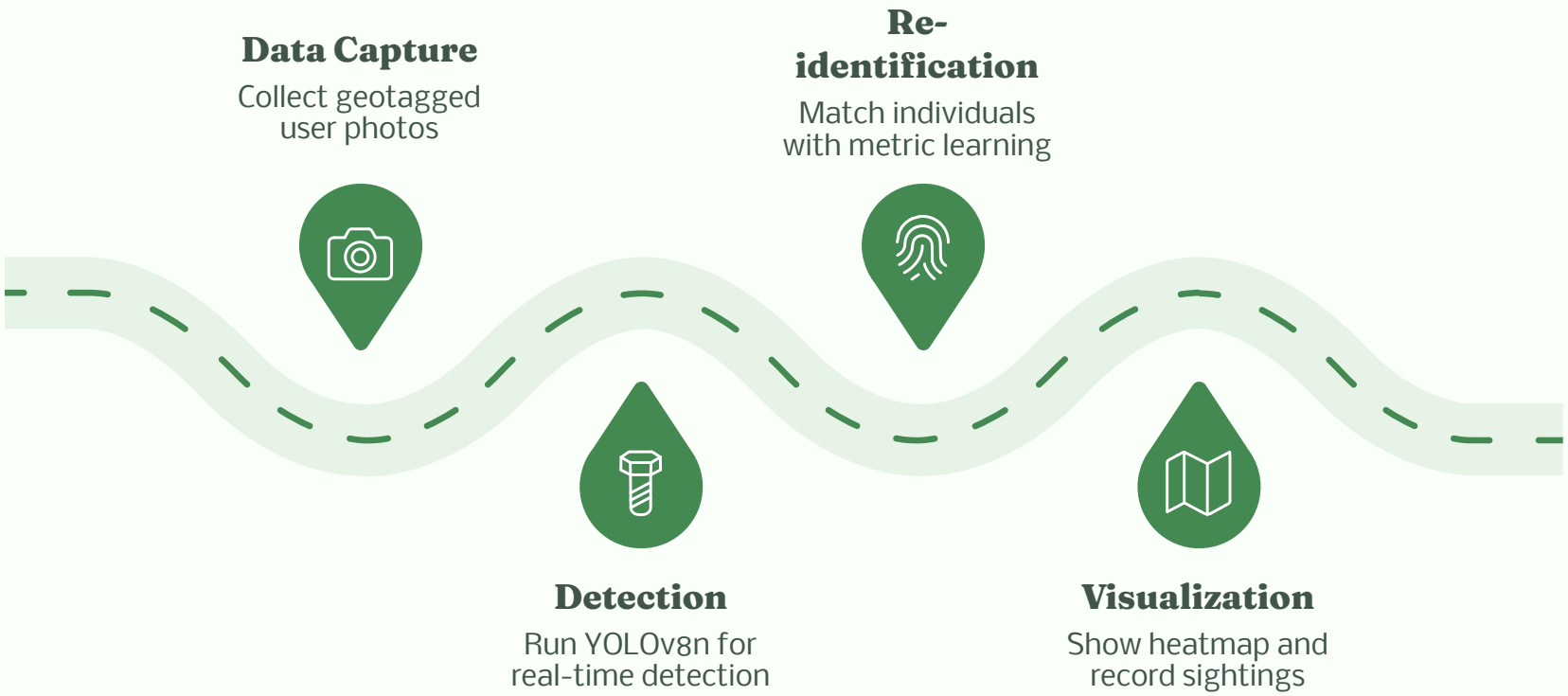
AI Pipeline



The on-device pipeline: user photo → YOLOv8n detect.tflite → crop bounding box → EfficientNet-B0 reid.tflite → cosine similarity → either match existing individual or register new identity.

Detection confidence threshold: 0.15 · Re-ID similarity threshold: 0.55 · Embedding update gate: 0.70 – all tuned empirically during on-campus testing.

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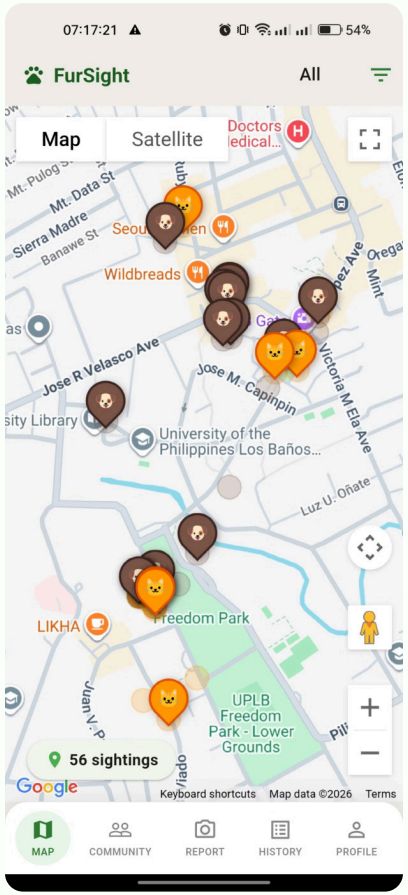
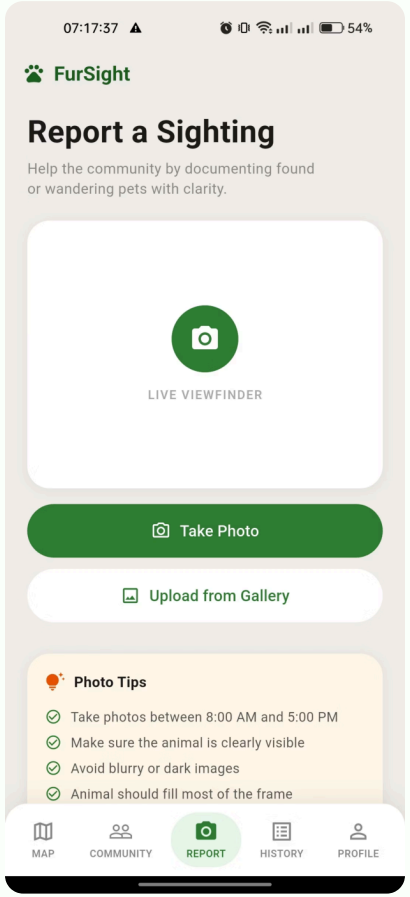
Key App Features

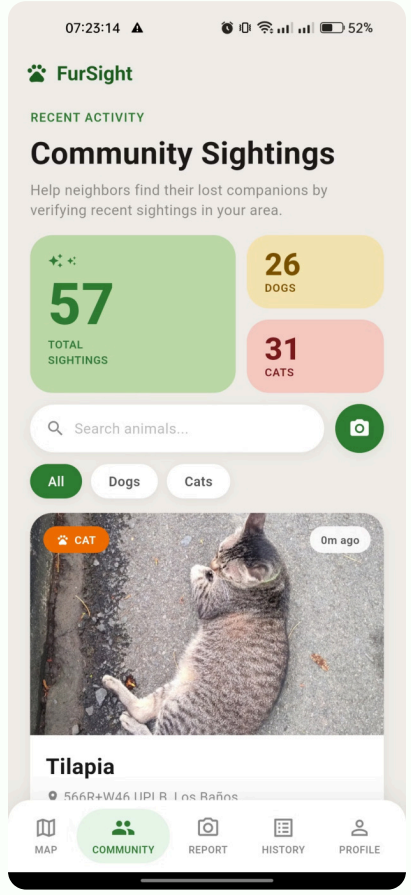
Heatmap



Report a Sighting



Community Sightings



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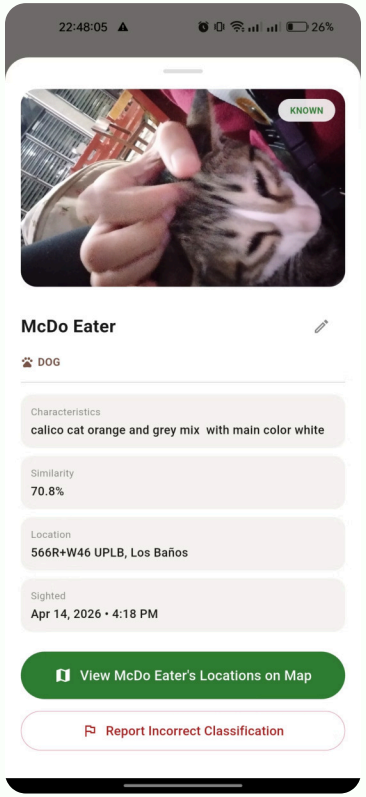
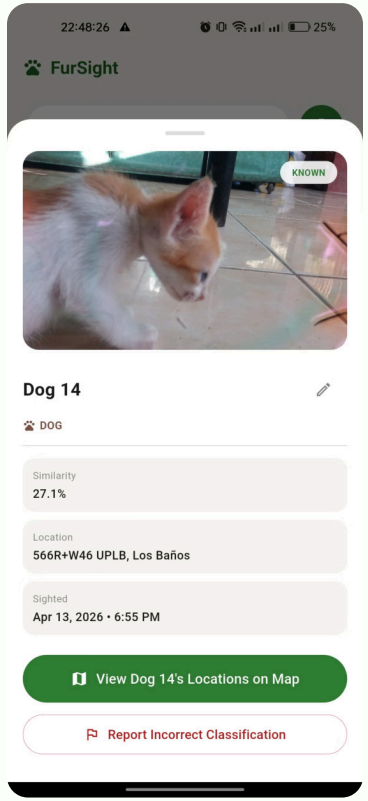
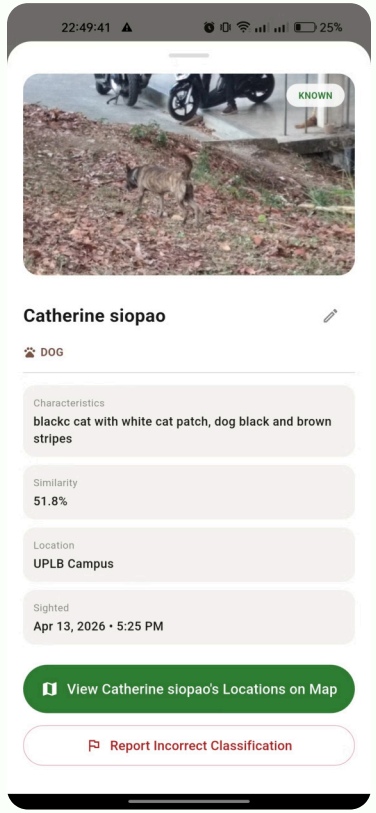
Common Errors & Misclassifications



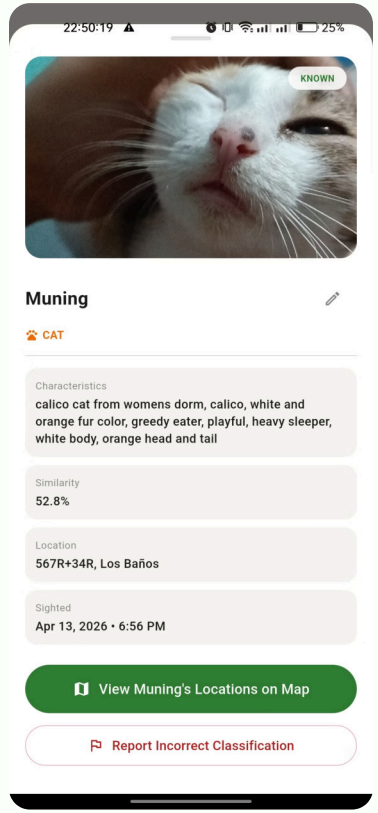
Cat misclassified as DOG – human hand occluding face and unusual top-down angle confused the model despite high similarity (70.8%)



Cat misclassified as DOG – head-down posture hides all facial features; very low similarity (27.1%) suggests weak match



DOG with wrong characteristics – small, distant animal camouflaged against leaf-covered ground; species and identity both wrong



Correct species, borderline re-ID – extreme close-up with no body context produced a weak embedding barely above the 0.55 threshold (52.8%)

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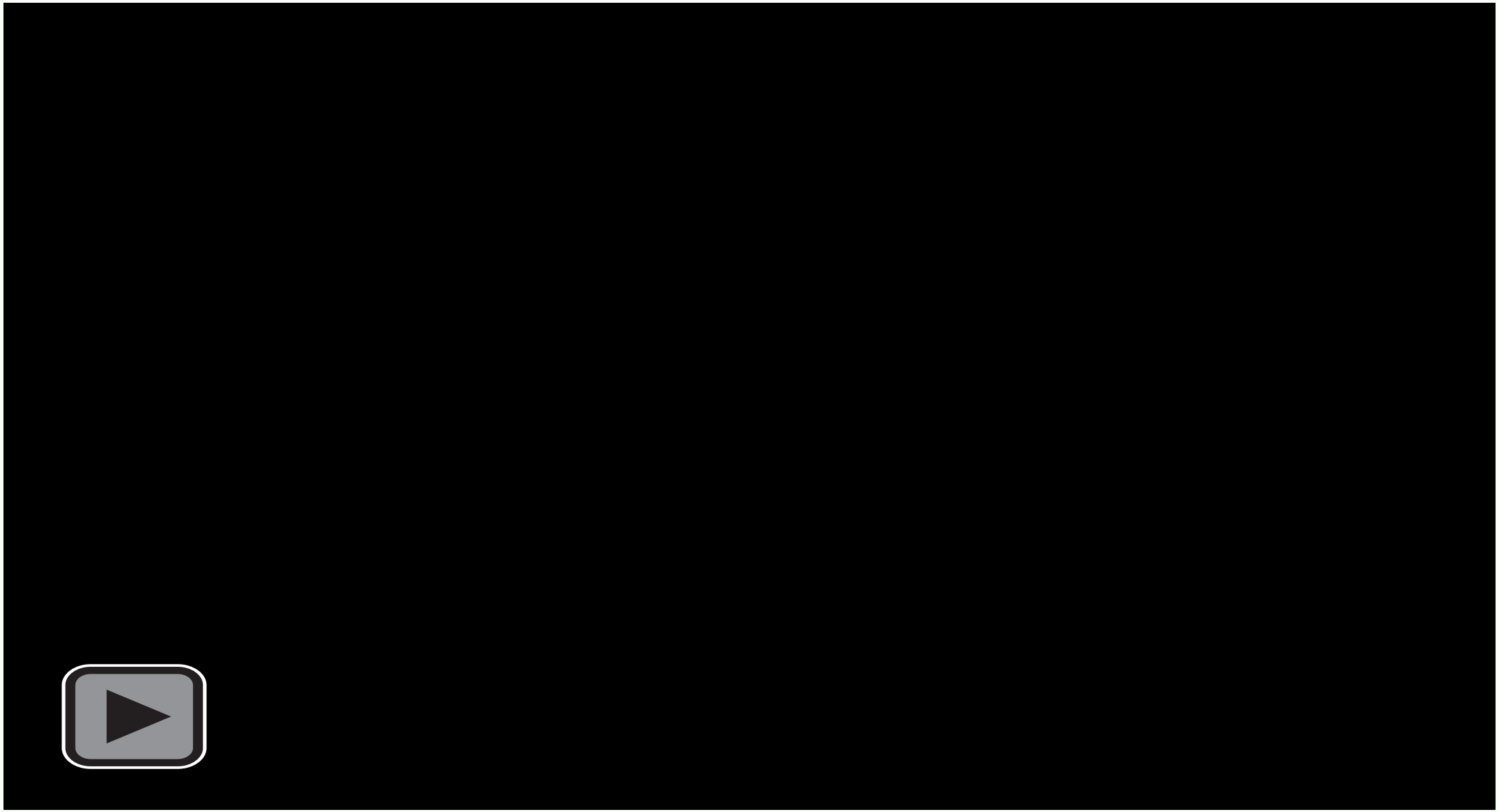

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Evaluation: Study Design

01

Instrument

- System Usability Scale (10 items, 5-point Likert).
- Supplemented by task difficulty ratings, AI accuracy ratings, and open-ended feedback.

02

Participants

- UPLB community.
- 20 students completed the evaluation.

03

Tasks

- Log in
- Submit sighting
- View feed
- Explore heatmap
- Check logs
- Find hotlines
- Read How It Works.

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Usability Results (SUS)

4.40

Learn Quickly

Most users reported fast learnability.

4.45

Confident Use

High confidence when using the app.

4.20

Ease of Use

App seen as easy and well-integrated.

1.20

Not Cumbersome

Low perceived workload overall.

> **Average SUS: 83.88 (Excellent)**

Overall ratings: 9 × Excellent · 11 × Good · Recommendation likelihood: 9.50/10

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AI Accuracy & User Feedback

Detection accuracy: 3.85/5 (cat vs. dog classification)

Re-ID accuracy: 3.50/5 – expectedly lower due to limited per-individual images.

Common Errors

Misclassification in poor lighting, flat postures, or blurry images; re-ID failures when database has few images per animal.

User-reported Incorrect IDs

Users reported at least one incorrect identification; retake frequency low. Top causes: lighting, blur, pose, distance.

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Conclusions & Future Work

Achievements

FurSight achieved a detection mAP@0.5 of 0.9944 and re-identification accuracy of 0.9401. Usability evaluation with 20 UPLB respondents yielded a mean SUS score of 83.88 (Excellent), confirming that the system is both technically sound and accessible to community users.

Future Work

Expand training dataset with campus-specific animals, redesign AI result display for clarity (confidence wording), add manual pin-drop location input, introduce a missing-animal reporting feature, and extend platform support beyond Android.

Impact

FurSight shows community-driven, AI-assisted monitoring is feasible and welcomed – a practical foundation for campus animal welfare and monitoring efforts.

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Thank you.

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